

# Structure of the protein subunits in the photosynthetic reaction centre of *Rhodopseudomonas viridis* at 3 Å resolution

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*The molecular structure of the photosynthetic reaction centre from Rhodopseudomonas viridis has been elucidated using X-ray crystallographic analysis. The central part of the complex consists of two subunits, L and M, each of which forms five membrane-spanning helices. We present the first description of the high-resolution structure of an integral membrane protein.*

The first steps in photosynthesis are mediated by protein-pigment complexes in the photosynthetic membrane. Light energy is absorbed primarily by light-collecting complexes and transferred to reaction centres where it is used with a very high quantum efficiency to transport electrons across the membrane. The reaction centre from the purple bacterium *Rhodopseudomonas viridis* consists of the four protein subunits L, M, H and c-type cytochrome<sup>1</sup> with four covalently linked haem groups<sup>2</sup>. Other prosthetic groups associated with the reaction centre are four bacteriochlorophyll *b* (BChl-*b*), two bacteriopheophytin *b* (BPh-*b*), two quinones, one non-haem ferrous iron and carotenoid(s)<sup>1</sup>. Apart from the cytochrome, which has been found as an integral constituent of reaction centre complexes only in some bacteria, it is generally accepted that reaction centres of purple bacteria are organized in a very similar way. This can be concluded from the subunit composition, from homology between the amino-acid sequences of the L and M subunits, from chromophore composition and from similar spectroscopic and electron-transfer properties (for reviews see, for example, refs 3–7).

The reaction centre from *Rps. viridis* was the first to be crystallized<sup>8</sup>; the photochemical activity of the crystalline material has been demonstrated<sup>9</sup>. An X-ray structure analysis of these well-ordered crystals allowed the calculation of an electron-density map at 3 Å resolution; an atomic model of the prosthetic groups was deduced from this map<sup>10</sup>. The arrangement of the pyrrole ring systems of BChl-*b* and BPh-*b* molecules in the central region of the reaction centre (assumed to consist of the subunits L and M) shows an approximate 2-fold symmetry with two closely associated BChl-*b*s near the symmetry axis. These two molecules form the 'special pair' postulated from the results of spectroscopic investigations<sup>11</sup>. The special pair is symmetrically in contact with two BChl-*b* molecules, each of which has a BPh-*b* as a neighbour. These chromophores form two branches that are apparently suitable as electron pathways. The right-hand side branch (see Fig. 4 in ref. 10) ends with a quinone close to the non-haem iron located on the local symmetry axis. The second quinone and the carotenoid are either not present or disordered in the crystal. The four haem of the cytochrome show a roughly linear arrangement with one haem near the special pair<sup>10</sup>.

This chromophore model correlates well with the electron-transfer steps determined from spectroscopy<sup>12</sup>. After absorption of light an electron is transferred from the excited primary electron donor (the special pair), possibly via a BChl-*b*<sup>13</sup> to an intermediate acceptor (BPh-*b*), and further to a primary quinone acceptor ( $Q_A$  or tightly bound quinone).  $Q_A$  reduces a secondary quinone acceptor ( $Q_B$  or loosely bound quinone).  $Q_B$ , when fully reduced and protonated, is commonly assumed to leave the reaction centre, and to be replaced from a quinone pool in the membrane<sup>14</sup>. The oxidized special pair is reduced by the cytochrome.

Here we report further interpretation of the electron-density map. We describe the folding of the polypeptide chains of the protein subunits in the reaction centre and the spatial relation between protein and prosthetic groups. With this structure we can present the first model of an integral membrane protein determined at nearly atomic resolution.

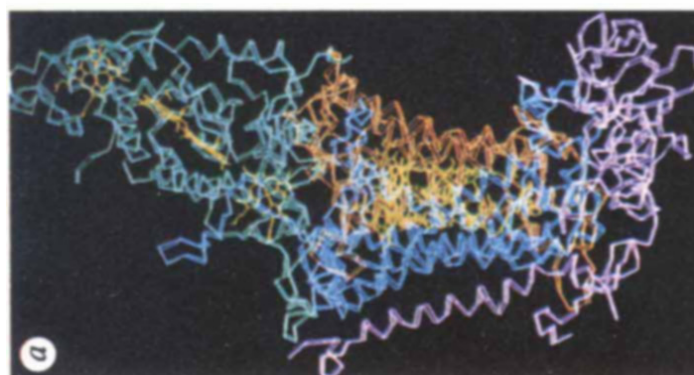
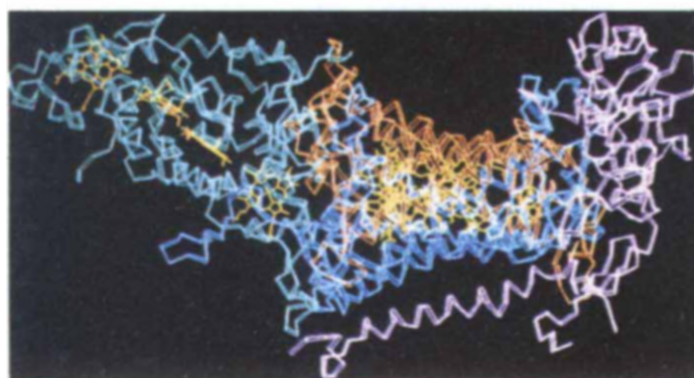
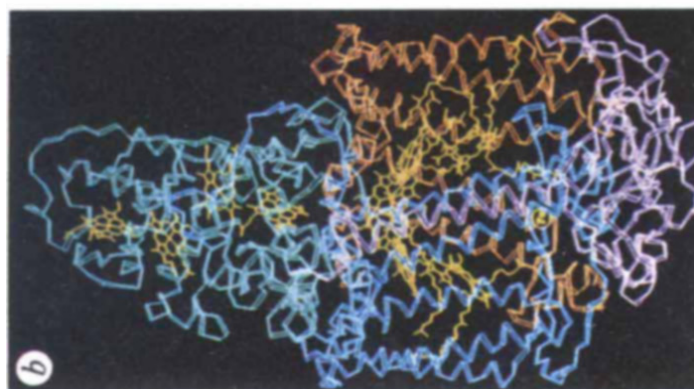
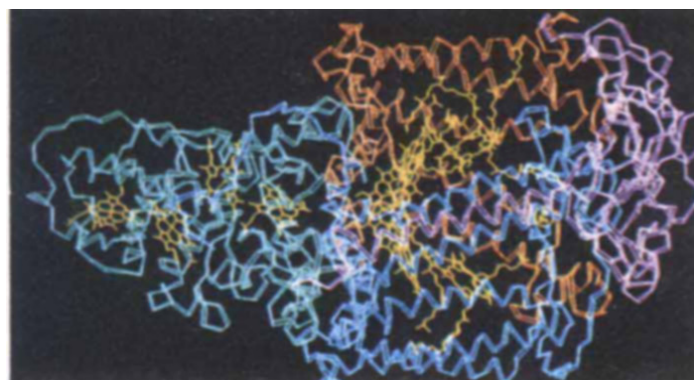
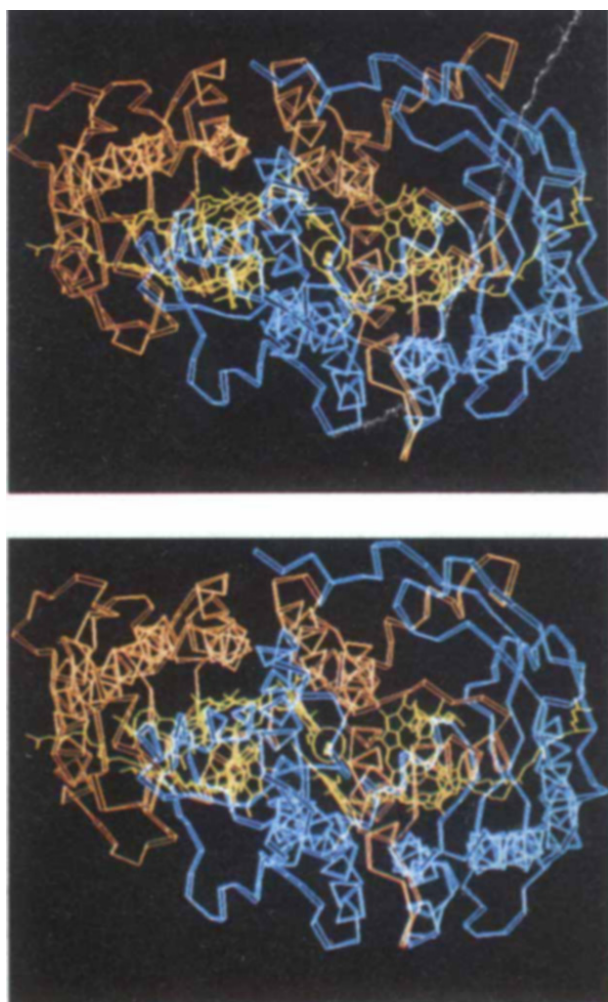
## Electron-density map

The reaction centre crystals belong to the space group  $P4_32_12$  with unit cell constants of  $a = b = 223.5$  Å,  $c = 113.6$  Å, and contain one centre per asymmetrical unit<sup>8,10</sup>. The electron-density map at 3 Å resolution was calculated with phases from multiple isomorphous replacement using five heavy-atom derivatives; this map was improved further by solvent flattening<sup>15</sup>. A detailed report on data collection, map calculation and arrangement of prosthetic groups is described elsewhere<sup>10</sup>.

The chromophore model facilitated the identification of the protein subunits of the reaction centre. The haem groups are evidently bound to the cytochrome subunit; BChl-*b*, BPh-*b*, non-haem iron and quinone are known to be associated with the subunits L and M in reaction centres from purple bacteria<sup>4,16,17</sup>. Available amino-acid sequences of 46, 30 and 28 residues<sup>18</sup> at the amino-termini of the subunits L, M and cytochrome, respectively, could be located in the map. The H subunit was identified on the basis of a partial gene sequence corresponding to 68 residues from the amino terminus.

For model building of the protein subunits a minimap (scale 2 mm Å<sup>-1</sup>), and interactive display systems (Vector General 3400 and Evans & Sutherland PS330) driven by the program FRODO<sup>19</sup> were used. Optimum fit of helices and large side chains to the density was achieved with the real-space refinement option in FRODO<sup>20</sup>. Segments with unknown amino-acid sequence were built as polyalanine. Residue numbers were prefixed with the letters L, M, C and H, respectively, for distinction between subunits.

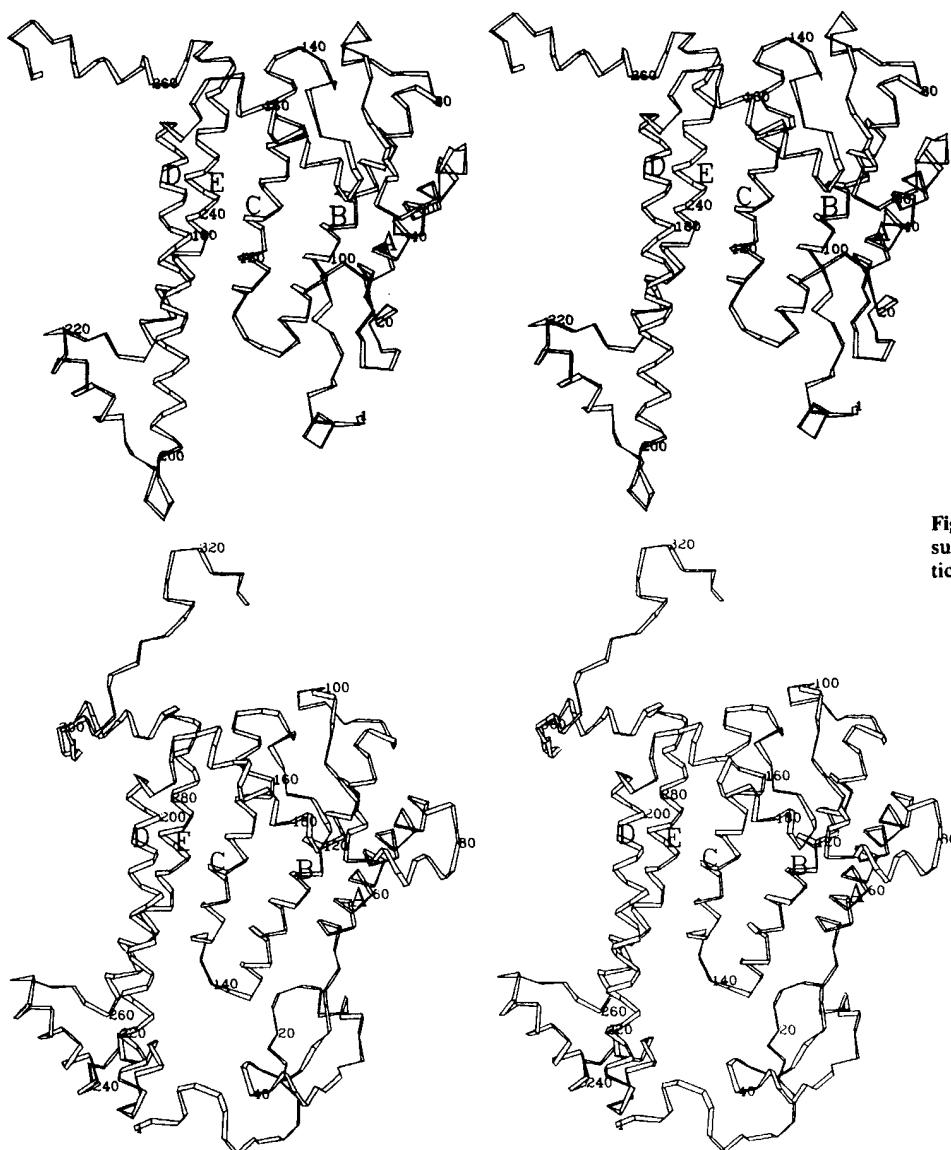
The good quality of the map allowed unambiguous chain tracing even in the absence of amino-acid sequence information. Recently the complete amino-acid sequences of the subunits H<sup>21</sup>, L and M (H.M., K. A. Weyer, H. Gruenberg, I. Dunger, D. Oesterhelt and F. Lottspeich, in preparation) became available from the corresponding gene sequences and were incorporated into the model. The amino-acid sequence of the cytochrome, presently being investigated with protein-chemical methods (K. A. Weyer, H.M. and F. Lottspeich, unpublished data), is still incomplete. However, peptide sequences resulting from this work could be located in the electron density and used for model building. Currently, these pieces add up to ~60% of the residues. The absence of a contiguous sequence for the cytochrome leads to an uncertainty in the residue numbering for this subunit caused by possible erroneous omission or insertion of residues during chain tracing.



**Fig. 1** (Parts *a*, *b* above.) Stereo drawing of the complete reaction centre complex in two orientations. Panel *a* is rotated with respect to *b* by 90° around a vertical axis. Green, cytochrome; orange, L; blue, M; purple, H; yellow, prosthetic groups. Figs 1-5 were produced by a computer program written by A. M. Lesk and K. D. Hardman<sup>50</sup>.

**Fig. 3** (Left) Stereo drawing of the L-M complex and associated chromophores viewed along the L-M local diad. Protein chains are represented as ribbons. Colours as in Fig. 1.





**Fig. 2** Stereo drawings of *a*, subunit L; *b*, subunit M represented as ribbons in identical orientations. Transmembrane helices A-E are labelled.

A crystallographic *R*-value ( $R = \sum ||F_{\text{obs}}| - |F_{\text{calc}}|| / \sum |F_{\text{obs}}|$ ) was calculated when the model contained ~70% of the correct sequence. For 46,658 reflections between 7- and 3-Å resolution we obtained  $R = 0.359$ ; this value is exceptionally low for a protein model before crystallographic refinement.

### The reaction centre complex

The course of the polypeptide chains and the location of the prosthetic groups of the centre are shown in Fig. 1. The central part consists of the subunits L and M which bind the BChl-*b*, BPh-*b*, the quinone and the non-haem iron. The cytochrome subunit is bound to a flat surface of L-M close to the special pair of BChl-*b*. A large contact area and the carboxy-terminal arm of the M subunit are responsible for the tight binding of the cytochrome to L-M. The major part of the H subunit is in contact with the other flat surface of the L-M complex that is close to the binding sites of the quinone and the non-haem iron. The amino-terminal segment of H forms a helix extending from the cytochrome side along the surface of the M subunit. The longest dimension of the reaction centre complex is ~130 Å between the tips of cytochrome and H subunit.

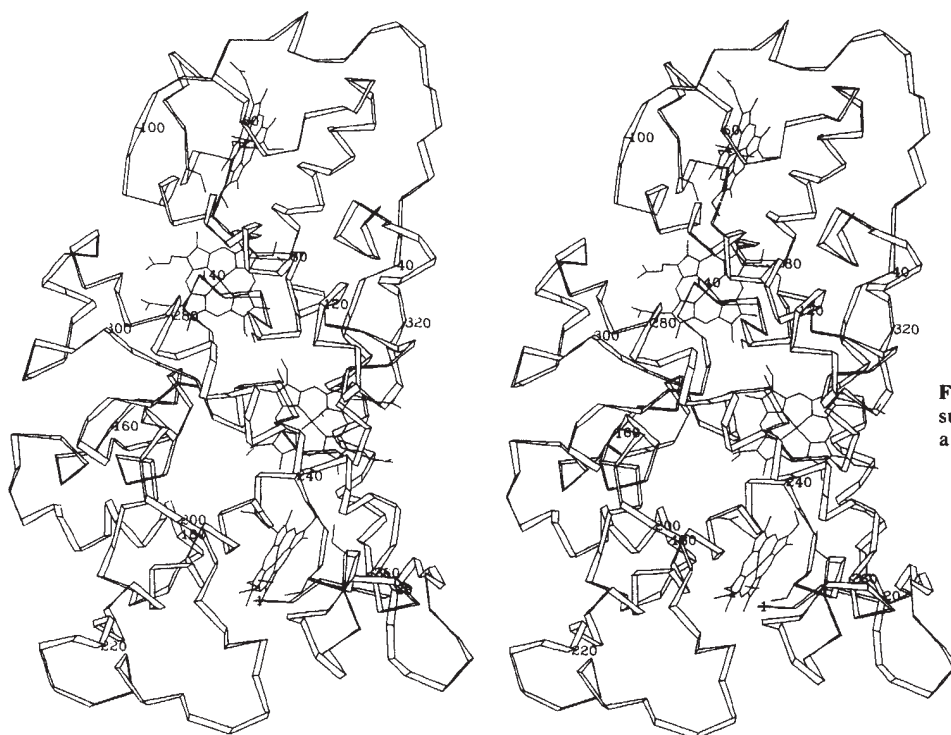
In our electron-density map there is no clear indication of the presence of ordered lipid or detergent that would provide direct evidence on the position and orientation of the membrane relative to the reaction centre complex. However, the highly hydrophobic surface of the central region of the centre indicates that this part (that is, the subunits L and M and the amino-

terminal helix of H) spans the membrane. The same conclusion follows from the fact that with electron transfer from the special pair to the quinone(s) a charge separation across the membrane takes place. Recently, the amino-terminus of the L-subunit was localized at the cytoplasmic side of the photosynthetic membrane from *Rhodospirillum rubrum*<sup>22</sup>. Assuming the same orientation for the reaction centre from *Rps. viridis*, this means that the cytochrome is located at the periplasmic side, and the major part of the H subunit is at the cytoplasmic side of the membrane.

### Subunits L and M

**Subunit structure.** Although L and M differ in length by 50 residues (L, 273 residues; M, 323 residues), their folding is generally very similar. Using the method of Rossmann and Argos<sup>23</sup>, 219 C $\alpha$ -positions of M could be rotated onto corresponding positions of L to an r.m.s. distance of 1.25 Å. The rotation axis is defined by the polar angles  $\phi = 43^\circ$ ,  $\psi = 39.4^\circ$ ,  $\kappa = -179.3^\circ$  (where  $\phi$  is the angle between the rotation axis and *y*-axis;  $\psi$  is the angle between projection of the rotation axis on *x-z* plane and *x*-axis; and  $\kappa$  is the rotation angle). At the present level of accuracy this L-M local diad is virtually identical to the local diad relating ring atoms of BChl-*b* and BPh-*b* ( $\phi = 43^\circ$ ,  $\psi = 37.6^\circ$ ,  $\kappa = -177.1^\circ$ )<sup>10</sup>. Figure 2 shows ribbon drawings of subunits L and M in the same orientation. Outstanding structural features of both subunits are five long helical regions (mostly  $\alpha$ -helix) numbered A-E (LA-LE in the L subunit.





**Fig. 5** Stereo drawing of the cytochrome subunit. The protein chain is represented as a ribbon; haem groups and linked cysteine side chains are shown as line drawings.

The non-haem iron is bound by five protein ligands, four histidines and one glutamic acid. The histidines are located in the transmembrane helices LD (L190), LE (L230), MD (M217) and ME (M264), and obey the L-M local 2-fold symmetry. They are arranged approximately in a plane perpendicular to the L-M local diad. The fifth iron ligand, Glu M232, sits on the local L-M diad. The structure near the non-haem iron is in good agreement with conclusions drawn from extended X-ray absorption fine-structure studies<sup>28,29</sup>.

As reported previously<sup>10</sup>, we believe that the quinone identified in the electron-density map is the tightly bound quinone  $Q_A$  (menaquinone-9 in *Rps. viridis*<sup>30</sup>).  $Q_A$  is located near  $BP_L$ ; its head group is bound exclusively to residues of M in the helix connection D-E with close contacts to His M217 (one of the iron ligands), to the peptide nitrogen of M258 and to Trp M250 whose side chain is nearly parallel to the  $Q_A$  head group. Its isoprenoid chain is mainly in contact with  $BP_L$  and with residues of L. It was shown that the loosely bound  $Q_B$  (ubiquinone-9 in *Rps. viridis*<sup>30</sup>) can be lost during purification of the reaction centres<sup>14</sup> or during crystallization<sup>30</sup>. To find the  $Q_B$  site, the binding of the competitive inhibitors for  $Q_B$  *o*-phenanthroline<sup>31</sup>, terbutryn (a herbicide)<sup>32</sup> and of ubiquinone-1 in the crystal was studied by soaking these compounds into the crystals (*o*-phenanthroline, 2 mM for 3 days; terbutryn, 1 mM for 3 days, followed by 0.01 mM for 3 days; ubiquinone-1, 1 mM for 3 days), collecting X-ray data and analysing these data using the difference-Fourier technique. These studies showed binding of the three compounds to a pocket formed near the H contact surface by parts of the transmembrane helices LD and LE, and by the LD-LE connecting segment. Additional binding sites were observed for *o*-phenanthroline (two), and ubiquinone-1 (seven). From these results we conclude that the single common binding site of the three compounds is the  $Q_B$  binding site. Only this site, which is related to the  $Q_A$  site by the L-M local diad, is close to the non-haem iron as required to explain the broadening of electron-spin resonance lines observed for  $Q_B^-$  (ref. 33). The  $Q_B$  site is in contact to the iron ligand His L190; the iron itself is located midway between  $Q_A$  and  $Q_B$ . This arrangement strongly suggests that the iron participates in electron transfer from  $Q_A$  to  $Q_B$ . However, experiments with iron-depleted reaction centres from *Rps. sphaeroides* showed that electron transfer is only moderately slower in the absence of the iron<sup>34</sup>.

The binding sites of  $Q_A$  and  $Q_B$  are in regions of L and M, where both subunits markedly deviate from the L-M local symmetry. Whereas  $Q_A$  is completely shielded from the H subunit by the M chain, the probable  $Q_B$  binding site is accessible from H through an opening in the H contact surface. These observations correlate the different binding of  $Q_A$  and  $Q_B$  to structural differences between L and M; they also suggest that one function of H is related to  $Q_B$  binding. A more detailed description of the reaction centre structure near the prosthetic groups will be given elsewhere.

### The H subunit

With 258 residues H is the smallest subunit of the reaction centre. The course of its polypeptide chain is shown in Fig. 4. The amino-terminus is near the cytochrome contact surface of L-M; residues H1, H2, H3 and H5 are in contact with residues from the cytochrome. Residues H12 to H37 form a helix which runs along the convex side of the surface formed by the transmembrane helices of M, with contacts to helices ME and MD (see Fig. 1). The amino-acid sequence in this helical region up to residue H32 is also primarily hydrophobic; the carboxy-end of the helix (H33-H37) and further on to H39 consists of charged residues (Arg-Arg-Glu-Asp-Arg-Arg-Glu) which interact partly among themselves, and probably also with the polar head groups of membrane lipids *in vivo*.

Altogether we find 11 transmembrane helices in the reaction centre complex, 6 running from the cytoplasmic side to the periplasmic side and 5 running in the opposite direction. The question of whether the dipole moments of these helices can facilitate electron transfer across the membrane, as proposed recently<sup>35</sup>, requires a detailed inspection of the model and will be discussed elsewhere.

The absence of significant electron density for residues H48 to H56 near the end of the transmembrane helix indicates that this segment is disordered in the crystal. The segment H57-H115 lies on the surface of L-M; residues H66-H70 and H74-H78 form a two-stranded antiparallel  $\beta$ -sheet. The remainder of H is folded into a globular domain. Residues H156-H120 form two layers of  $\beta$ -sheet which enclose a hydrophobic core. Both sheets consist of three strands and show a right-handed twist. In sheet 1 the strands (H162-H174, H179-H187 and H191-H197) are antiparallel; in sheet 2 the middle strand (H207-H210)



is accompanied by an antiparallel strand (H200–H204) and a parallel strand (H156–H159). Near the carboxy-terminus residues H231–H249 form a helix that runs parallel to the H contact surface of L–M.

Little is known about the function of H in the reaction centre complex. Crosslinking studies<sup>16</sup> indicate that H is close to the light-collecting complexes in the membrane; this observation suggests that H has a role in determining the relative orientation of light-collecting complexes and reaction centre. As mentioned above, H may also play a role in  $Q_B$  binding—residue H177 (Glu) and its neighbours have access to the  $Q_B$  binding pocket in subunit L. They could be involved in donating protons to the photochemically reduced  $Q_B$ . This is in accord with experimental results on L–M complexes of *Rps. sphaeroides*<sup>36</sup> which show abnormal electron transfer from  $Q_A$  to  $Q_B$  in the absence of H.

## The cytochrome

The present model of the cytochrome subunit contains 333 residues. Figure 5 shows the course of the polypeptide chain and the location of the haem groups. Near the amino-terminus, residues C7–C11 and C20–C23 form an antiparallel  $\beta$ -sheet that lies on the L half of the cytochrome contact surface. From there the chain proceeds to the tip of the subunit (around C50); residues C25–C37 form a helix. In the following bulk of the subunit four recurring structural motifs can be observed: a helix runs parallel to the ring plane of each haem group (residues C67–C82, C102–C119, C222–C240 and C262–C286, respectively); the haems are covalently attached near the carboxy termini of these helices where the chains turn back towards the haems. The electron density strongly indicates that these covalent links are thioether bridges, and that the sequences of the haem binding segments are Cys–X–X–Cys–His as in other c-type cytochromes; the His residues are the fifth ligands to the haem irons. The haems are numbered according to the order of attachment to the protein; for haem 1 the first covalent link is to residue C87, haem 2, C132, haem 3, C244 and haem 4, C305. All haem irons appear to be six-coordinated; the sixth ligands to the irons of haem 1, 2 and 3 are located in the helices associated with the haems (residues C74, C110 and C233, respectively). The sixth ligand to the haem 4 iron (C124) is in the segment between the helix parallel to haem 2 and the haem 2 attachment site. According to the electron density the sixth ligands to the haem irons could be histidines or methionines.

As reported previously<sup>10</sup>, the haem groups are arranged with an approximate 2-fold symmetry: pyrrole ring atoms of haems 3 and 4 can be rotated onto those of haems 1 and 2 with polar angles  $\phi = 117.2^\circ$ ,  $\psi = -70.2^\circ$  and  $\kappa = -176.5^\circ$  to an r.m.s. distance of 0.41 Å. Haems 1 and 2 and haems 3 and 4 are related by improper rotations ( $\sim 105^\circ$ ) with large screw components ( $\sim 12$  Å). Using the same method as for the L–M comparison<sup>23</sup>, 66  $C^\alpha$  positions from the region C226–C315 could be rotated onto corresponding  $C^\alpha$  positions from between C67 and C142 to an r.m.s. distance of 1.1 Å. The polar angles of this rotation are  $\phi = 116.7^\circ$ ,  $\psi = -68.9^\circ$  and  $\kappa = -177.8^\circ$ , very close to those for the haem rotation. This internal 2-fold symmetry includes about one third of the cytochrome subunit. More extended structural homologies with 2-fold symmetry in protein molecules have been reported for rhodanese<sup>37</sup> and  $\gamma$ -crystallin<sup>38</sup>. The intramolecular symmetry is broken in the segment connecting the binding sites of haem 2 and 3; this segment contributes to the contact of cytochrome to L–M. The carboxy-terminal segment ends near the helix associated with haem 2. Haem 1 is accessible from the environment at its pyrrole rings I and IV; haem 2 appears to be almost completely buried; and haems 3 and 4 are accessible at their propionic acid groups. Pyrrole rings I and IV of haem 3 are closest to the special pair of BChl-*b* in the L–M complex. Apparently, protein folding in the cytochrome subunit of the reaction centre is different from both cytochromes C2 and cytochromes C3 with known three-dimensional structure (see ref. 39 for review).

## Structure of related proteins

**Reaction centres from other purple bacteria.** Amino-acid sequences are known for the reaction centre subunits L, M and H from *Rps. capsulata*<sup>40</sup>, and for L and M from *Rps. sphaeroides*<sup>41,42</sup>. The sequences of L and M from *Rps. viridis* show between 50 and 60% homology with those of the corresponding subunits from the other two bacteria. On the basis of hydrophobicity five transmembrane helices each were postulated<sup>40–42</sup> in segments of L and M homologous with the transmembrane helix regions in our model. All the ligands to the BChl and to the non-haem iron described above are also conserved, and the polypeptide segments connecting these ligands have the same length in all three organisms. These observations clearly indicate that the structures of L and M from *Rps. capsulata* and *Rps. sphaeroides*, and the arrangement of BChls and non-haem iron are very similar to those found for *Rps. viridis*. Significant differences in polypeptide chain length occur exclusively at the carboxy-termini where L from *Rps. viridis* is shorter by 8 residues, and M is longer by 17 residues. These additional residues of M are in contact with the cytochrome in *Rps. viridis*: the reaction centres of *Rps. capsulata* and *Rps. sphaeroides* lack a permanently bound cytochrome subunit.

For the H-subunits from *Rps. capsulata* and *Rps. viridis* the sequence homology is 37% (ref. 21). Common features of these two H-subunits include a hydrophobic region near the amino terminus which corresponds to the transmembrane helix of H in our model. In addition, many of the residues involved in the contact between H and L–M are conserved in both organisms, so that structural similarity can also be expected between the H subunits from *Rps. capsulata* and *Rps. viridis*.

**Photosystem II.** Recently, amino-acid sequences of polypeptides from photosystem II reaction centres<sup>43–47</sup> have been reported. Sequence homologies were found between L and M of purple bacteria and the D1 polypeptide (also referred to as  $Q_B$  protein or herbicide binding polypeptide) from photosystem II of higher plants<sup>40,42</sup> and between D1 and D2 (ref. 45) which is also a constituent of photosystem II. In these plant proteins, histidines which correspond to the ligands of the special pair, and to those of the non-haem iron in our model are conserved. Even a Trp residue corresponding to M250, which forms part of the  $Q_A$  binding site in reaction centres of purple bacteria, can be found in the D2 proteins. These observations corroborate our recent proposal<sup>48</sup> that proteins D1 and D2 form the core of the photosystem II reaction centre, similar to the subunits L and M in purple bacteria. Based on functional and sequence homology we believe that D1 and D2 span the membrane five times, contrary to a proposed seven-helix structure<sup>49</sup>, where the helix connections A–B and D–E were probably misinterpreted as transmembrane helices.

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# Identification of the signal molecules produced by wounded plant cells that activate T-DNA transfer in *Agrobacterium tumefaciens*

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*We show here that Agrobacterium tumefaciens virulence (Vir) gene expression is activated specifically by the plant molecules acetosyringone (AS) and  $\alpha$ -hydroxyacetosyringone (OH-AS). These molecules induce the entire vir regulon in Agrobacterium as well as the formation of T-DNA intermediate molecules. AS and OH-AS occur specifically in exudates of wounded and metabolically active plant cells and probably allow Agrobacterium to recognize susceptible cells in nature.*

SOIL bacteria often form specialized interactions with plant cells. Usually a particular bacterium interacts with only a few species and/or types of plant cells<sup>1,2</sup>. A primary step in the formation of a bacterial/plant interaction is the detection by the bacterium of the appropriate susceptible plant cell. This recognition then triggers the activation of the bacterial genes whose products direct the development and/or maintenance of the interaction. The soil is a complex biological and chemical environment, and the signals that mediate the detection of a specific plant cell by a bacterium are unknown.

*A. tumefaciens*, a soil phytopathogen, genetically transforms dicotyledonous plant cells to cause the neoplastic disease crown gall<sup>3,4</sup>. Only cells that have been wounded are seen to be susceptible<sup>5,6</sup>. The bacterium transfers a specific segment of DNA, the T-DNA, from its large (> 200 kilobases, kb) tumour-inducing (Ti) plasmid to the susceptible plant cell, where it becomes integrated into the nuclear genome<sup>7,8</sup>. In the Ti plasmid, the T-DNA is defined and bounded by identical 25-base pair (bp) direct repeats; only DNA between these T-DNA borders is transferred to the plant genome<sup>9-12</sup>. During co-cultivation with plant cells, independent T-DNA circles are formed in *Agrobacterium*; these molecules arise by a specific recombination between the 25-bp sequences at the ends of the T-DNA and are potential intermediates in the transfer of the T-DNA from *Agrobacterium* to the plant cell<sup>13</sup>.

The Ti plasmid genes required for plant transformation are not contained in the T-DNA but are located in the ~40-kb *vir* region<sup>14-16</sup>. Genetic analysis of the *vir* region of the A6 Ti plasmid has shown that it encodes at least six separate complementation groups, *virA*, *B*, *C*, *D*, *E* and *G*, and *pinF* that are organized as a single regulon (ref. 17; S.E.S., in preparation). In the vegetative bacterium only *virA* and *virG*, the *vir* regulatory genes, are significantly expressed; however, when *Agrobacterium* is co-cultivated with plant cells the expression of *virB*, *C*, *D*, *E*,

*G* and *pinF* is induced to high levels<sup>17,18</sup>. This activation of *vir* expression by plant cells probably initiates the steps of T-DNA transfer and integration into the plant cell genome. We have shown that *vir* induction and the production of T-DNA circles is mediated by a small diffusible factor produced by actively growing plant cells<sup>17</sup>. Here, we purify and establish the chemical identity of this factor, and demonstrate that its production is related to plant cell wounding and that its recognition by the bacterium is a highly specific process. This factor is likely to be the signal that allows *Agrobacterium* to recognize in nature a plant cell susceptible to transformation.

## Purification of signal molecules

To purify the plant molecule(s) that specifically signals *Agrobacterium* to initiate its interaction with plant cells requires first, an efficient and quantitative bioassay for a plant event in this transformation process, specifically the induction of *vir* gene expression; and second, a starting source of the signal molecule(s). We have previously described an assay for *vir* induction in *Agrobacterium* that used gene fusions between the pTiA6 *vir* loci and the *Escherichia coli lacZ* gene<sup>17,18</sup>. Briefly, in a bacterium carrying a *vir::lac* gene fusion, the production of  $\beta$ -galactosidase (the *lacZ* gene product) is controlled by the *vir* locus to which *lacZ* has been fused. Thus, the state of expression of the locus can be monitored by measuring the  $\beta$ -galactosidase activity present in the bacterium; increased activity indicates increased *vir* expression. (The relative amount of induced activity reflects the relative amount of *vir*-inducing activity to which the cell has been exposed.) Here we use *Agrobacterium* strain A348(pSM30) (ref. 17) to detect and measure *vir*-inducing activity. This strain contains wild-type pTiA6 and pSM30, a *virB::lac* fusion plasmid, and gives high levels (up to ~100-fold of basal activity) of induced  $\beta$ -galactosidase activity<sup>17</sup>.